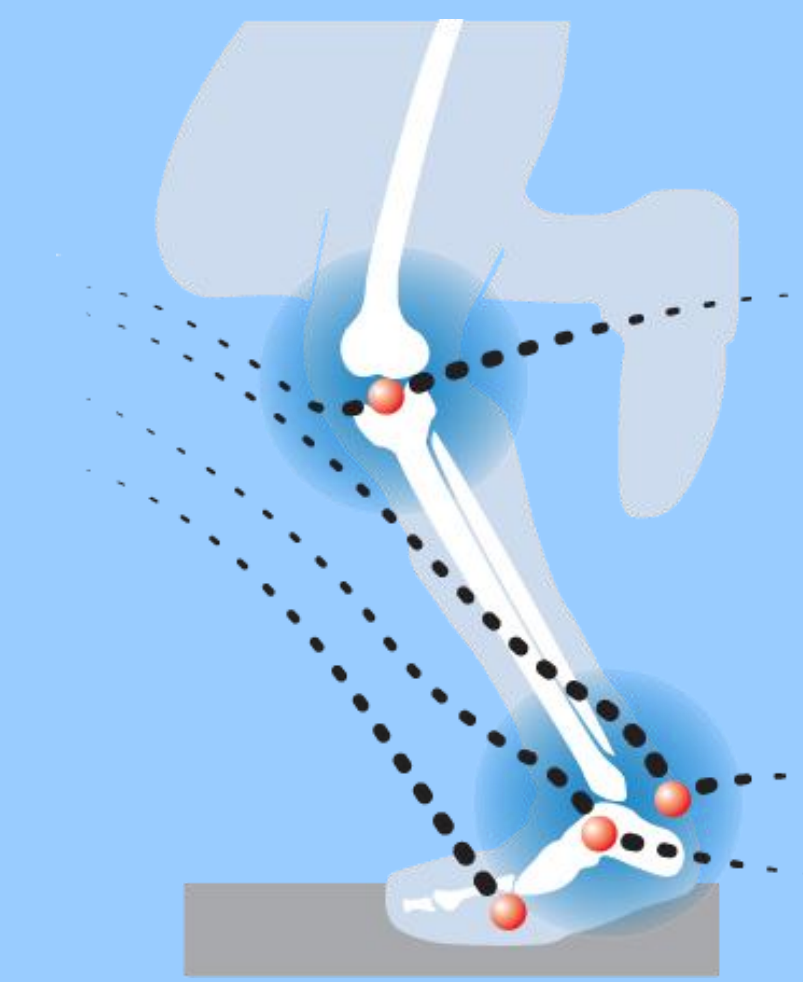


Comparisons of the Joint Responses to Surgical Transection and Traumatic Rupture of the ACL in a Rabbit Model

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Introduction

- Long-term participation in vigorous physical activity increases the risk of acute and chronic injuries to the knee.
- Two specific types of injury strongly associated with subsequent knee osteoarthritis (OA) are anterior cruciate ligament (ACL) and meniscal tears.
- Axial compressive loading of the knee produces an anterior shift of the tibia that result in isolated rupture of the ACL.¹
- These studies have shown acute damages in the articular cartilage and underlying subchondral and trabecular bone.²
- In order to study the chronic effects of an ACL rupture, surgical transection models of the ACL have been developed.³
- However, these models do not replicate the acute injury seen clinically to articular cartilage and subchondral bone.

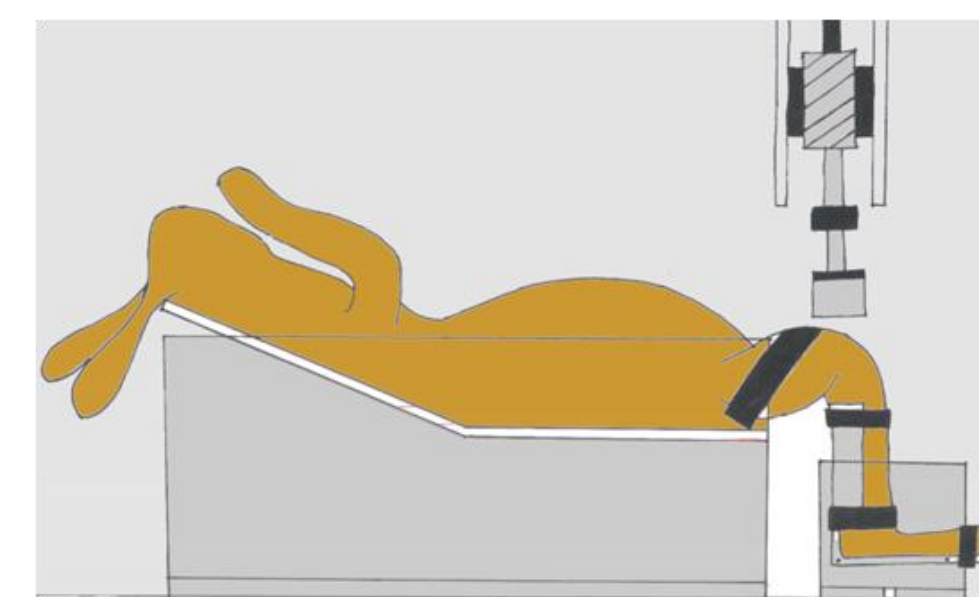
Hypothesis

Traumatic ACL rupture will result in more cartilage degeneration and bone remodeling of the impacted knee than surgical transection.

Methods

Impact ~ ACL Failure (ACLF)

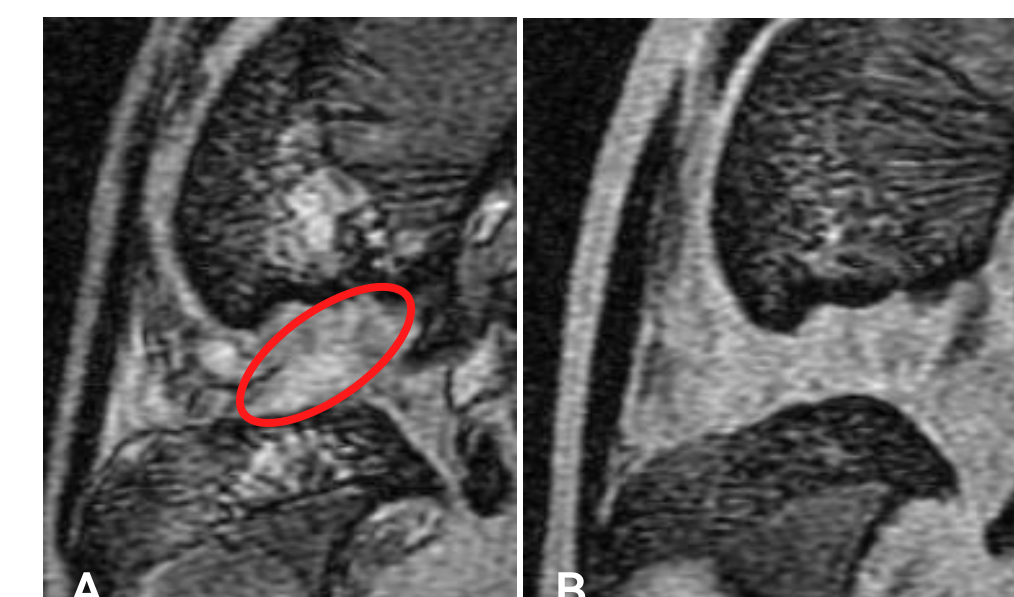
Three skeletally mature Flemish Giant rabbits were subjected to impact-induced ACL rupture. The right knee was not disrupted. All animals were sacrificed after 12 weeks and the joints were analyzed.



Schematic of loading device to apply 13J impact to the left, flexed knee.⁴

Surgery ~ ACL Transection (ACLT)

Three skeletally mature Flemish Giant rabbits underwent uni-lateral, surgical transection of the ACL. The right knee was not disrupted. All animals were sacrificed after 12 weeks and the joints were analyzed.

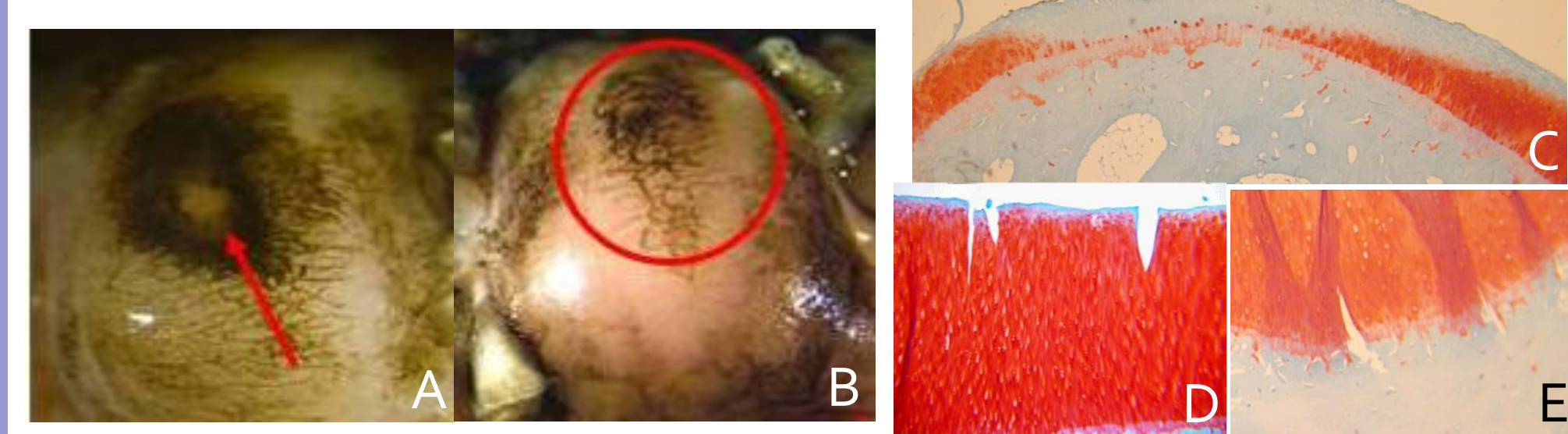


MRI showing normal control ACL (A) and no ACL after surgical transection (B).

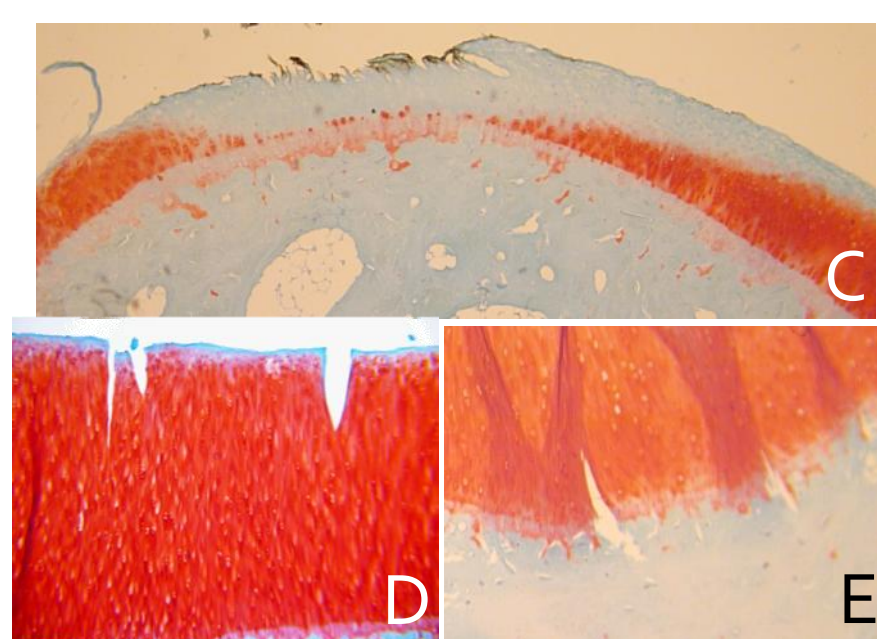
This investigation was approved by an All-University Committee on Animal Use and Care.

Animals received pain medication for 72 hrs following the ACL disruption, and were monitored regularly by a licensed veterinary technician.

Morphology and Histology

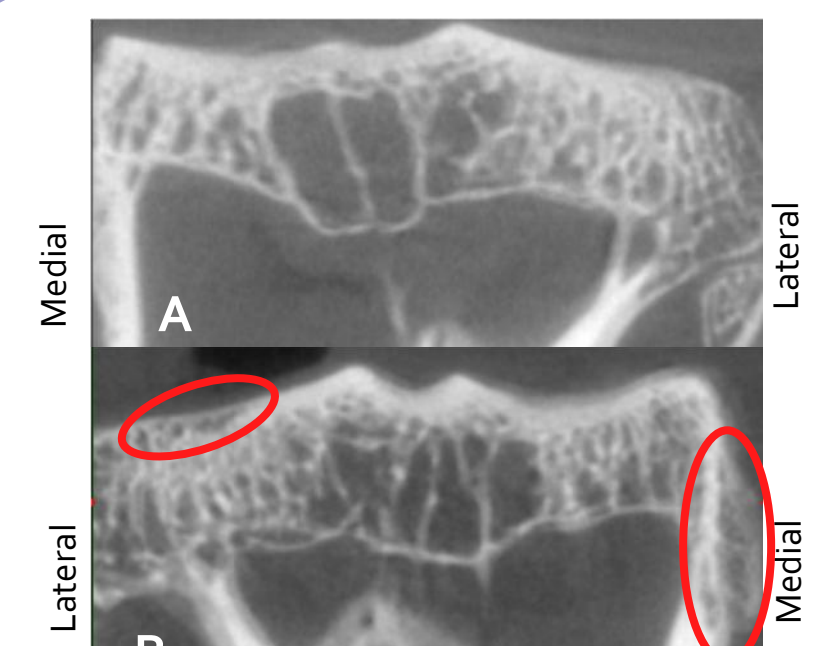


Medial femoral condyles showing full thickness ulceration of articular cartilage in the ACLF animals (A), but only fibrillation in the ACLT animals (B).

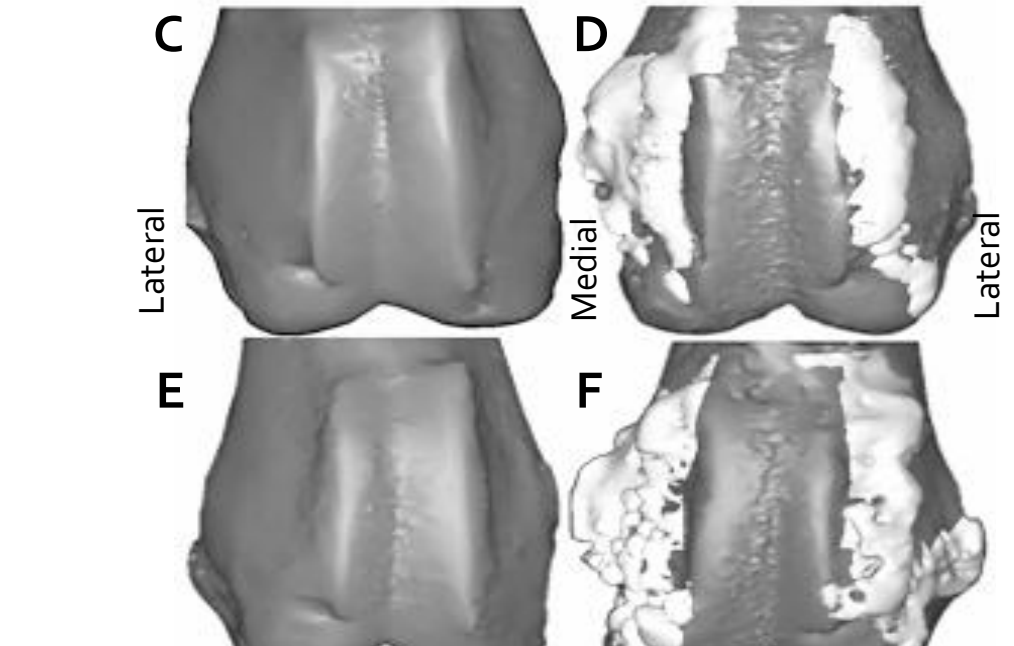


Surface erosion (C), cartilage fissures (D) and occult microcracks at the cartilage/subchondral bone interface (E).

MicroCT

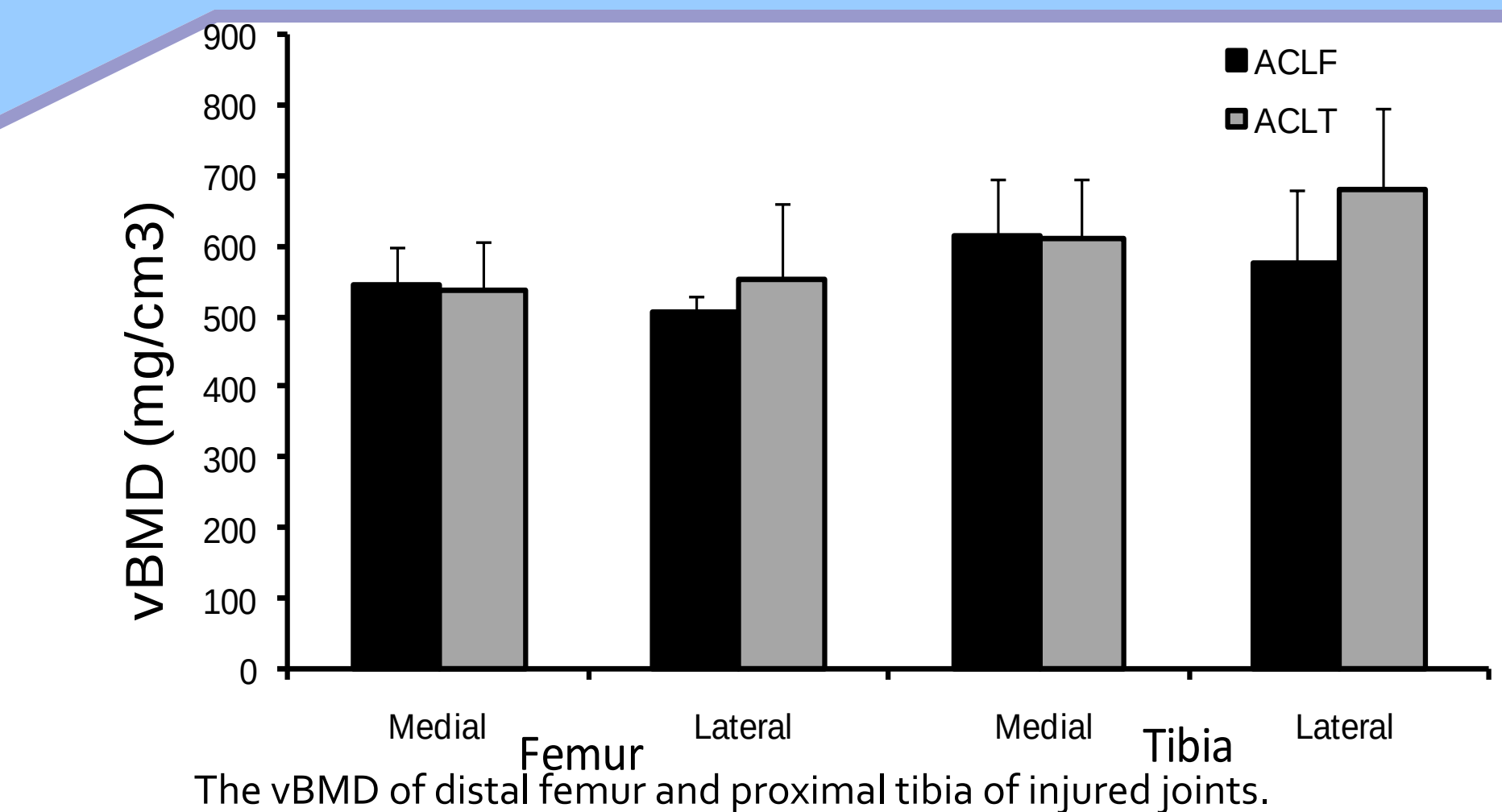
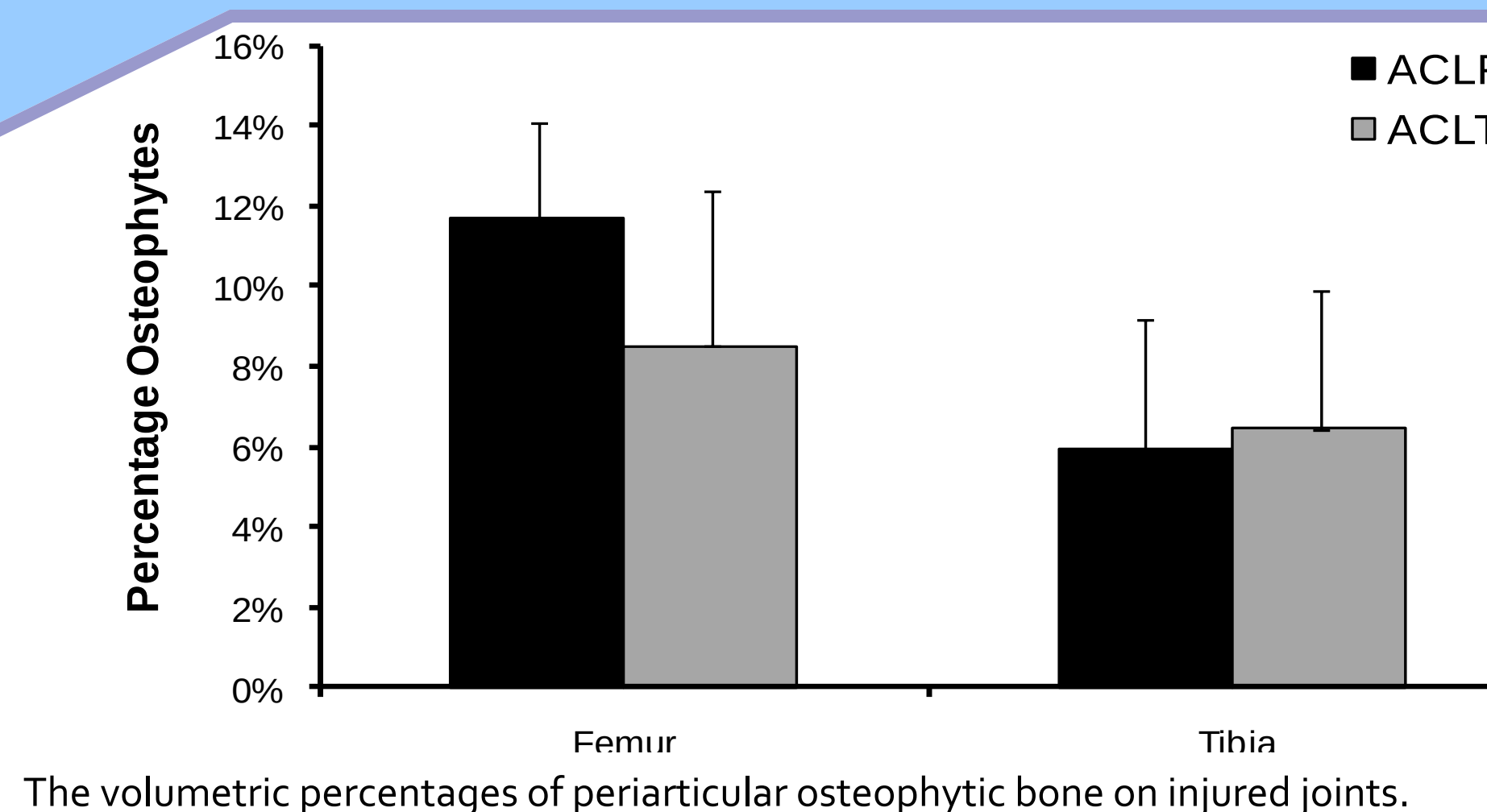
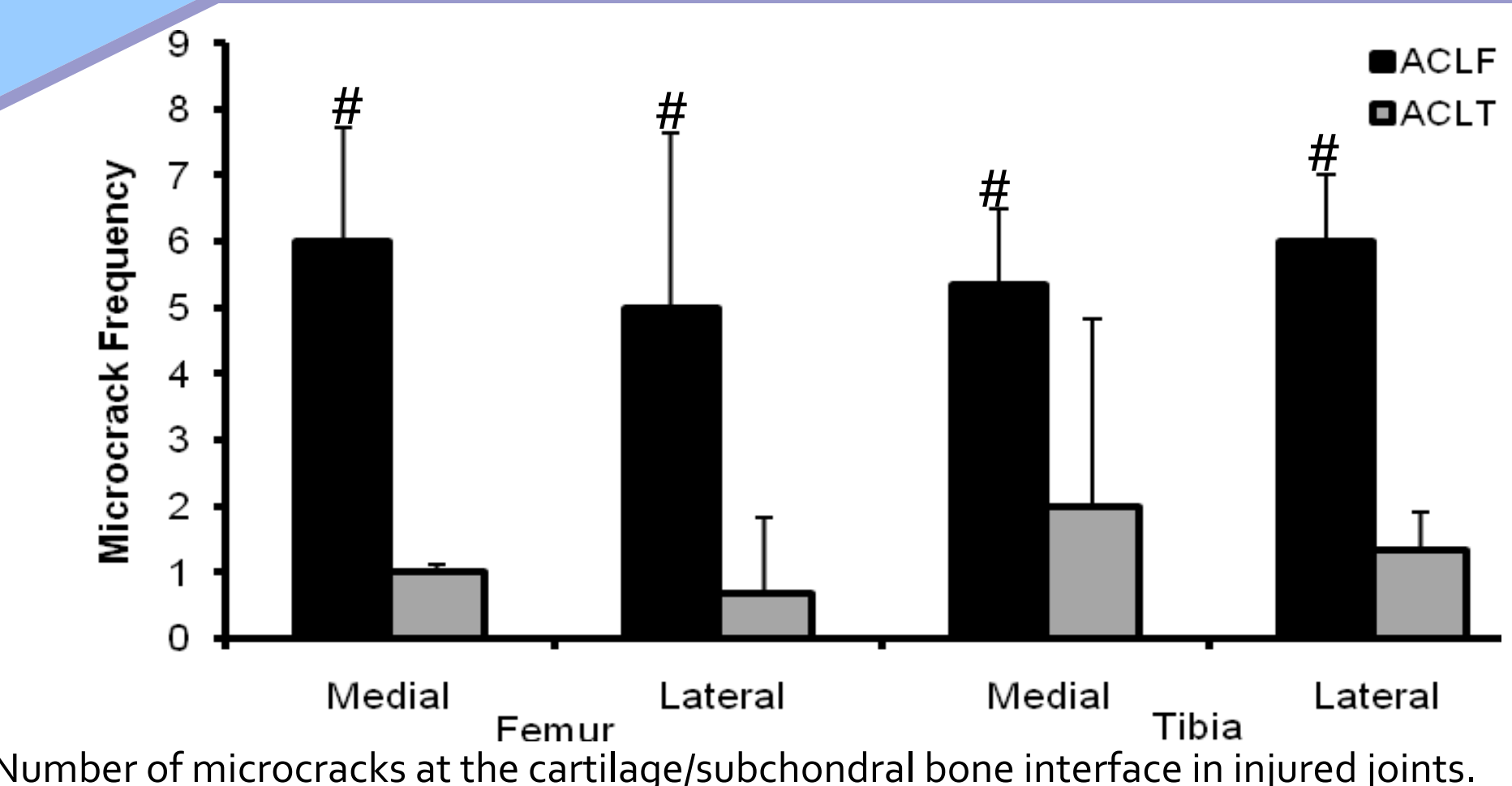
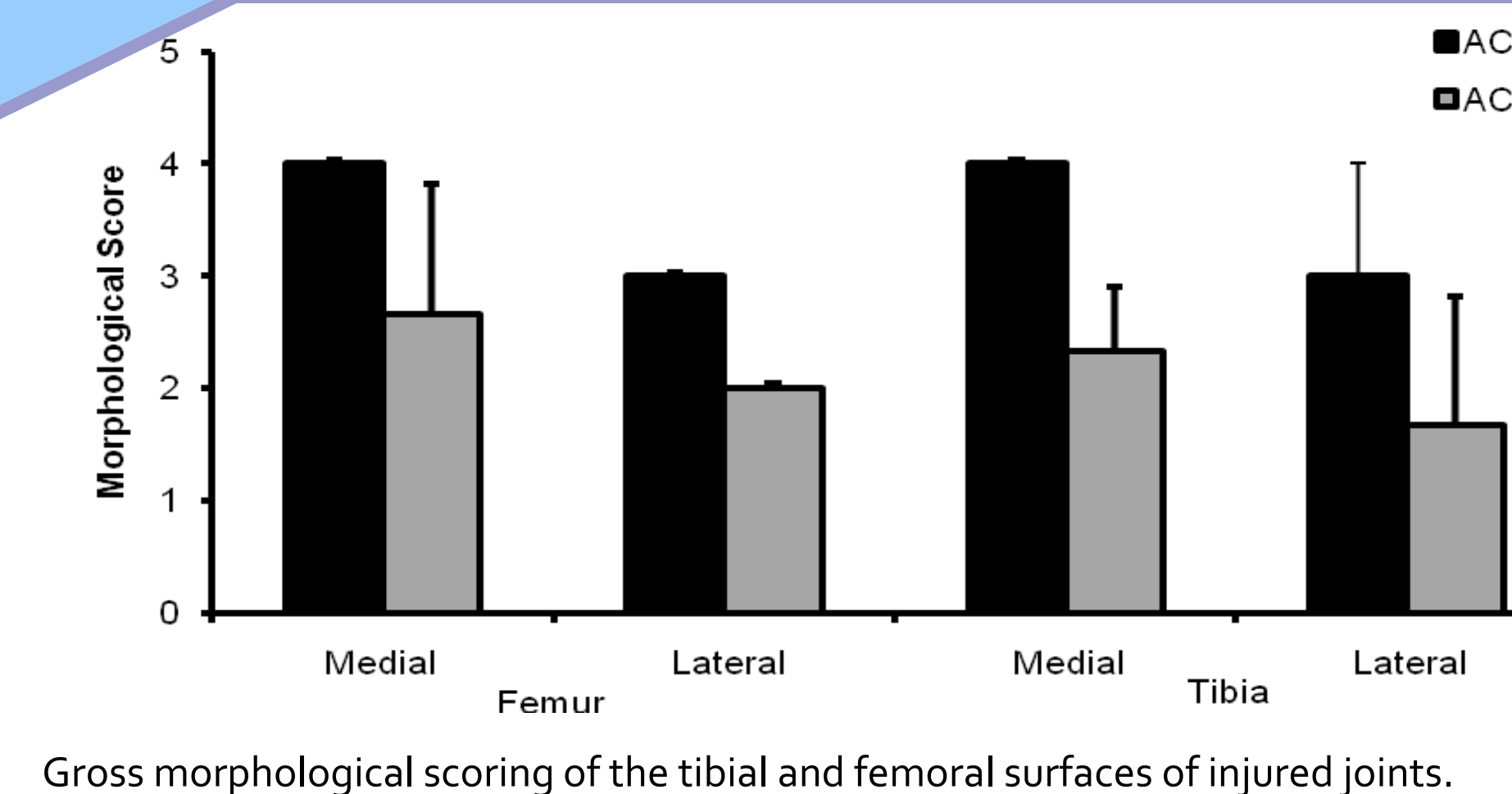


Normal control tibia (A) and osteophytes and subchondral bone remodeling in ACL-deficient tibia (B).



Highlighted osteophytes of control femurs (C: ACLF, E: ACLT) and ACL-deficient femurs (D: ACLF, F: ACLT).

Results



Discussion

- The conventional ACLT model and the novel ACLF model showed different degrees of cartilage degeneration and bone remodeling after 12 weeks, which may be due to acute damage to articular cartilage and subchondral bone.
- While acute osteochondral damage is documented for the same level of impact joint compression in a constrained tibia model, the level of joint degeneration is much lower without ACL rupture.⁵
- Increased joint degeneration in the ACLF group may also be due acute damage to the meniscus that is not included in ACLT group, or in the constrained tibia impact model.⁶
- Interestingly, acute damage to the meniscus for ACLF animals was located in the posterior lateral region, which correlates with the vBMD changes in the lateral compartment.

Conclusion

Including acute osteochondral and meniscal damage in experimental animal models may help develop new intervention strategies to limit or mitigate the high prevalence of post-traumatic OA.

- References:** (1) Meyer & Haut, J Biomech, 2008;41:3377-83. (2) Meyer et al., Am J Sports Med, 2008;36:1966-77. (3) Batiste et al., Osteo Cart, 2004;12:986-96. (4) Isaac et al., J Biomech. Eng, 2010;132(064501):1-4. (5) Isaac et al, J Biomech, 2010;43:1682-88. (6) Killian et al, J Surg Res, 2010;164:234-41.